Total questions: 35
Instructions: Answer all questions. For scenario questions, explain your diagnosis and the steps or commands you would use.

Section A: Multiple-Choice Questions — 10

Choose one answer for each question.

1. An EC2 instance's status checks show **System status check failed**. What is the most likely cause?
    - A. The operating system has a full disk
    - B. A problem with the underlying AWS host
    - C. The security group blocks SSH
    - D. The instance has insufficient IAM permissions
2. Which Linux command shows free and used space on mounted filesystems in a human-readable format?
    - A. du -sh
    - B. df -h
    - C. lsblk -f
    - D. findmnt -D
3. A process is consuming too much CPU. Which command sends it a signal that lets it clean up before exiting?
    - A. kill -9 <PID>
    - B. kill -TERM <PID>
    - C. kill -STOP <PID>
    - D. kill -CONT <PID>
4. Which EC2 purchasing option is generally suitable for a fault-tolerant workload that can tolerate interruptions?
    - A. On-Demand Instances
    - B. Reserved Instances
    - C. Spot Instances
    - D. Dedicated Hosts

5. A Linux file has permissions 640. What access does the group have?
    - A. Read and write
    - B. Read only
    - C. Write only
    - D. No access
6. Which command displays processes interactively, including CPU and memory usage?
    - A. top
    - B. pwd
    - C. mount
    - D. chmod
7. An application on an EC2 instance needs temporary credentials to access an AWS service. What is the recommended approach?
    - A. Store long-lived access keys in a script
    - B. Add credentials to the AMI
    - C. Attach an IAM role to the instance
    - D. Store credentials in a text file
8. Which command searches recursively for ERROR in files under /var/log?
    - A. grep -R "ERROR" /var/log
    - B. find /var/log "ERROR"
    - C. cat -R "ERROR" /var/log
    - D. locate -r "ERROR" /var/log
9. Which EC2 storage option is temporary and is lost when the instance is stopped or terminated?
    - A. EBS volume
    - B. EBS snapshot
    - C. Instance store
    - D. AMI
10. Which command shows the current status and recent log information

for a systemd service?

- A. systemctl status <service>
- B. chmod status <service>
- C. ps status <service>
- D. mount status <service>

Section B: Scenario-Based Questions – 15

For each scenario, describe the likely cause, how you would investigate it, and the steps you would take to resolve it.

1. An EC2 instance was reachable over SSH yesterday but times out today. Its state is **running**. Explain how you would check its public IP, security group rules, instance status checks, and SSH service.
2. A web server responds to curl localhost on the instance, but external users cannot reach it. Explain how you would check whether the application is listening on the correct interface and port, and inspect the host firewall and security group.
3. A Linux server reports "No space left on device," but df -h shows free disk space. Explain how to check inode usage and identify which directories or files may be responsible.
4. An EC2 instance becomes unresponsive during traffic spikes. Explain how you would distinguish CPU, memory, disk I/O, and network bottlenecks using Linux commands and AWS monitoring.
5. A process ignores a normal termination request. Explain how you would identify the process, inspect its state, send progressively stronger signals, and avoid terminating the wrong process.
6. An EC2 instance cannot download operating system updates. Explain how you would check DNS resolution, connectivity, proxy settings, firewall rules, and the package manager's logs.
7. A newly attached EBS volume appears in lsblk but cannot be used by an application. Explain how you would check, format if appropriate, mount, and persist the filesystem across reboots.
8. A user can log in to a Linux server but cannot read a file created by another user. Explain how you would inspect ownership, permissions, groups, and ACLs, then fix access safely.
9. After a reboot, a service that used to start automatically is not running.

Explain how you would inspect its status, logs, and startup configuration using systemctl.

10. An EC2 instance's CPU credit balance frequently reaches zero, and application response times rise. Explain what this suggests and give two ways to address it.
11. A team wants to deploy a new application version while minimizing downtime and keeping a quick rollback option. Describe an EC2-based deployment approach and how you would verify application health before directing users to it.
12. A Linux server has a high load average, but CPU utilization is low. Explain what this may indicate and how you would investigate blocked or I/O-waiting processes.
13. An application needs to access an AWS service, but the team has placed access keys in a file on the instance. Explain the security risk and how you would replace this with a safer approach.
14. A root EBS volume is nearly full. Explain how you would identify what is consuming space, expand the volume safely, and make the additional capacity usable by the operating system.
15. SSH access works from most locations but fails intermittently from one office network. Explain how you would investigate the office's public IP, security group rules, host firewall, SSH logs, and possible connection limits.

Section C: Theoretical Questions – 10

Answer in your own words. Include examples or command syntax where useful.

1. Describe what an AMI, an EBS snapshot, and an EBS volume are. Explain how they can be used for instance recovery.
2. Explain the difference between stopping and terminating an EC2 instance. Include the typical effects on EBS volumes and instance-store data.
3. Explain how Linux file permissions work for the owner, group, and others. What do chmod 750 and chown user:group file do?
4. Explain the difference between a process and a service in Linux. Describe how systemd manages services and how to inspect a service's logs.

5. Describe how Linux identifies disks and filesystems. Explain the roles of lsblk, blkid, mount, and /etc/fstab.
6. Explain the difference between SIGTERM and SIGKILL. Why should SIGTERM generally be tried first?
7. Describe EC2 instance metadata and explain how an instance role provides temporary AWS credentials to software running on the instance.
8. Explain the difference between EBS and instance-store storage. Give a suitable use case for each.
9. Describe how you would investigate a Linux server experiencing high memory usage. Include at least three relevant commands or tools and what each reveals.
10. Explain how Linux handles disk space, inodes, and deleted files that are still held open by a running process. How would you investigate each issue?

10 More Questions: ELB and Auto Scaling

1. **MCQ:** Which ELB type is best suited for routing HTTP and HTTPS traffic based on URL paths or hostnames?
   A. Network Load Balancer
   B. Application Load Balancer
   C. Gateway Load Balancer
   D. Classic Load Balancer
2. **MCQ:** What does an Auto Scaling group do when an instance fails an EC2 health check?
   A. Restarts the instance automatically
   B. Replaces the unhealthy instance, depending on its health-check configuration
   C. Moves the instance to another account
   D. Stops scaling until an administrator intervenes
3. **MCQ:** What is the purpose of an ELB target group?
   A. To store AMI backups
   B. To register targets and run health checks
   C. To set an instance's CPU limits
   D. To create launch templates
4. **Scenario:** Users report slow responses during peak hours. An

Application Load Balancer sends traffic to an Auto Scaling group. Explain how you would check target health, scaling policies, instance metrics, and the group's minimum, maximum, and desired capacity.

5. **Scenario:** New instances launch successfully in an Auto Scaling group but receive no user traffic. Describe how you would investigate target group registration, health-check settings, application readiness, and the group's attachment to the load balancer.
6. **Scenario:** An Auto Scaling group repeatedly replaces instances even though the application appears to be working. Explain how you would investigate the ELB health-check path, expected response code, timeout, and instance startup time.
7. **Scenario:** A scheduled event is expected to increase capacity overnight, but the Auto Scaling group remains unchanged. What settings, schedule time zone, desired capacity, and scaling activity history would you check?
8. **Theory:** Compare **dynamic scaling** and **scheduled scaling**. Give an example of when each approach is appropriate.
9. **Theory:** Explain the purpose of an ELB health check. What could happen if the health-check path or success codes are configured incorrectly?
10. **Scenario:** You need to deploy a new application version with minimal downtime using an ELB and Auto Scaling group. Describe a safe rollout and rollback approach.